

JIAHAO TIAN

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🌐 [jhao1122.github.io](https://github.com/jhao1122)

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Education

School of Mathematical Sciences, Zhejiang University

Aug. 2023 – Jun. 2027 (Expected)

Bachelor of Science in Statistics

Hangzhou, Zhejiang, China

- **GPA:** 88.96/100 (4.01/4.30)
- **Current Coursework:** Advanced Probability Theory (Grad), Advanced Mathematical Statistics (Grad), Differential Manifolds and Riemannian Geometry (Grad), and Abstract Algebra.

Selected Coursework

Probability Theory (92)

Mathematical Statistics (95)

Stochastic Processes (99)

Real Analysis (87)

Functional Analysis (92)

Generalized Functions (95)

Regression Analysis (96)

Asymptotic Statistics (PhD, 96)

Stochastic Differential Equations (Grad, 96)

Research Experience

Heavy-Tailed Robustness of Multiple-Knockoff Aggregation Methods

Sep. 2025 – Present

Undergraduate Researcher | Advised by Prof. Lijun Wang

Hangzhou, China

- Conducted a systematic robustness study of e-value-, voting-, p-value-, and data-splitting-based knockoff aggregation methods under Gaussian, moderately heavy-tailed, and Cauchy regimes; identified severe power loss under extreme heavy tails even when single-run knockoffs remained effective.
- Characterized a **zero-dilution mechanism** in which Cauchy-driven instability produces severely truncated single-run selections, causing occasional signal discoveries to be diluted below the final aggregation threshold across repeated knockoff runs.
- Developed an asymptotic explanation using **regular variation**, exact null sign symmetry, concentration inequalities, and **Extreme Value Theory**; established multiplicity-driven threshold inflation and a vanishing upper bound on signal-selection probabilities in sparse high-dimensional regimes.
- Investigated robust feature statistics and efficient internal-threshold calibration through a compute-once, evaluate-many procedure for multiple-knockoff aggregation.
- **First-author manuscript in preparation.**

Distributional Balancing Weights for Conformal Calibration under Covariate Shift

Mar. 2026 – Present

Research Assistant | Advised by Prof. Guanhua Chen (UW-Madison)

Remote

- Developed a **distributionally balanced conformal calibration** framework under covariate shift, using response-free weights to align labeled source calibration covariates with an unlabeled target sample and estimate the target-domain conformity-score distribution without pointwise density-ratio estimation.
- Checked the technical conditions behind a general score-distribution and approximate coverage-error bound, including residual imbalance, function-class approximation, finite target-sample uncertainty, and calibration-weight concentration through **effective sample size (ESS)**.
- Specialized the framework to **characteristic function distance (CFD)** balancing; used Fourier analysis to connect spectral decay with score-function complexity, and implemented the method for **Split CP**, **CQR**, **UACQR-S**, and **UACQR-P**.
- **First-author manuscript in preparation.**

Academic Writing & Presentations

Academic Notes Repository | [jhao1122.github.io/Notes](https://github.com/jhao1122/Notes)

Spring 2026 – Present

- Authored technical notes and seminar materials on stochastic differential equations, real and functional analysis, asymptotic statistics, large deviations, and derandomized knockoffs.

Convergence of Random Series and Large Deviations

Fall 2025

Advanced Probability Theory Seminar, Zhejiang University

Hangzhou, China

Robustness of Multiple Knockoff Methods

Fall 2025

Weekly Statistics Seminar, Zhejiang University

Hangzhou, China

Title of the Talk

Fall 2026

Topics in Stochastic Analysis Seminar, Zhejiang University

Hangzhou, China

Teaching Experience

Teaching Assistant, Probability and Mathematical Statistics

Sep. 2026 – Present

School of Mathematical Sciences, Zhejiang University

Hangzhou, China

- Conducted office hours, graded assignments, and answered student questions for an undergraduate probability and mathematical statistics course with over 200 students.

Skills & Tests

Programming: Python, R, Shell, C/C++, Julia, MATLAB
Tools: Linux, Git/GitHub, L^AT_EX, Quarto, Typst, MkDocs
Standardized Tests: GRE Quantitative: 170/170 | TOEFL iBT: 110/120 (5.5/6.0)